



Scuola Superiore
Sant'Anna

Master of Science in Bionics Engineering

Academic Year 2025-2026

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Bioinspired and Soft Robotics

Soft Robotics Technologies

Traditional Technologies

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Classification

Actuator			Feature
<i>Electromechanical</i>			
DC motor	Wound field	Separately excited	Speed can be controlled either by the voltage across the armature winding or by varying the field current
		Shunt series	Constant-speed application High starting torque, high acceleration torque, high speeds with light load
		Compound	Low starting torque, good speed regulation Instability at heavy loads
	Permanent magnet	Conventional PM motor	High efficiency, high peak power, and fast response
		Moving-coil PM motor	Higher efficiency and lower inductance than conventional DC motor
		Torque motor	Designed to run for long periods in stalled or a low rpm condition
	Electronic commutation (brushless motor)		Fast response High efficiency, often exceeding 75% Long life, high reliability, no maintenance needed Low radio frequency interference and noise production
AC motor		AC induction motor	The most commonly used motor in industry Simple, rugged, and inexpensive
		AC synchronous motor	Rotor rotates at synchronous speed Very high efficiency over a wide range of speeds and loads Need additional system to start
		Universal motor	Can operate in DC or AC Very high horsepower per pound ratio Relatively short operating life
Stepper motor		Hybrid	Change electrical pulses into mechanical movement Provide accurate positioning without feedback
		Variable reluctance	Low maintenance



Classification

Actuator		Feature
<i>Electromagnetic</i>		
Solenoid type devices Electromagnets, relay		Large force, short duration On/off control
<i>Hydraulic and Pneumatic</i>		
Cylinder		Suitable for linear movement
Hydraulic motor	Gear type Vane type	Wide speed range High horsepower output
Air motor	Rotary type Reciprocating	No electric shock hazard Low maintenance
Valves	Directional control valves	



Electromechanical actuators

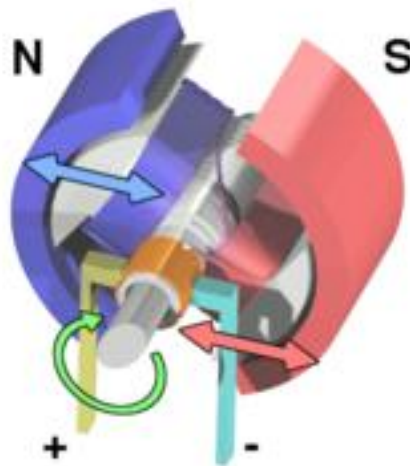
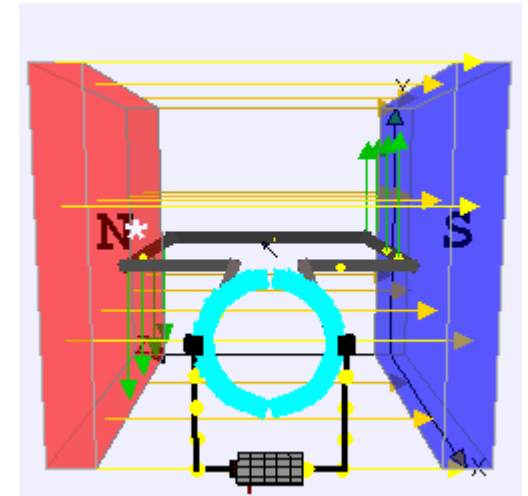
- The most common electromechanical actuator is a motor that converts electrical energy to mechanical motion.
- Motors are the principal means of converting electrical energy into mechanical energy in industry.
- Motors: DC motors, stepper motors, and AC motors.



Electromechanical actuators

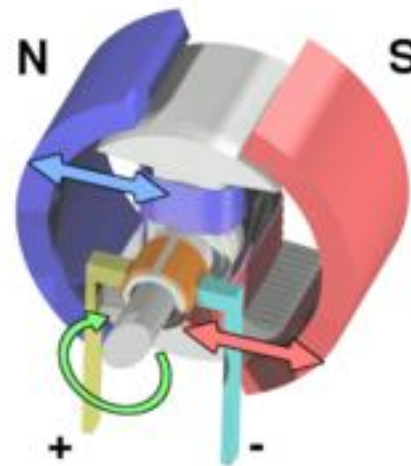
DC motors

Speed proportional to Voltage. The modern DC motor was invented by accident in 1873, when Zénobe Gramme connected the dynamo he had invented to a second similar unit, driving it as a motor.



A simple DC motor. When the current flows in the winding, all around the rotor a magnetic field is generated. The left side of the rotor is repelled by the left magnet and attracted by the right magnet.

The same applies to the bottom right part. The torque generates the rotation.



When the rotor will be horizontally aligned the commutator inverts the current direction flow, by inverting the magnetic field and the second part will begin.



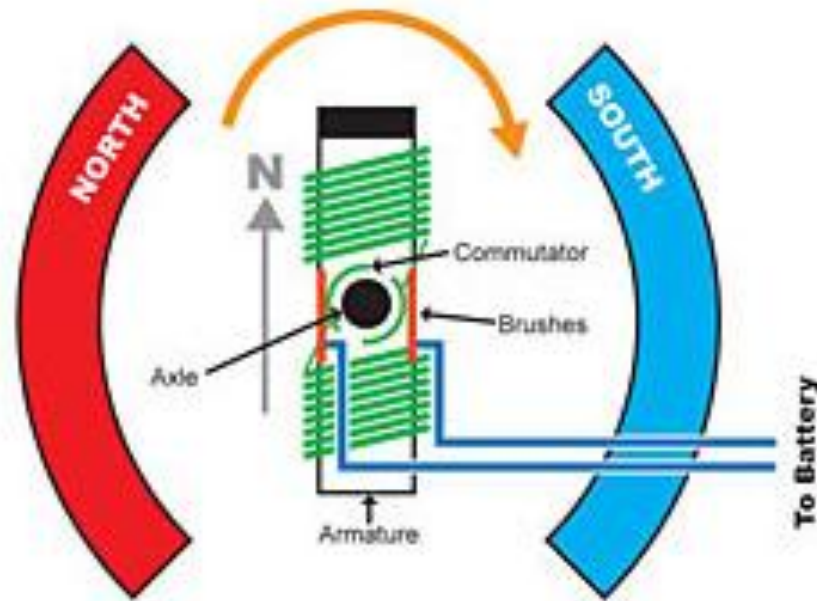
Electromechanical actuators

DC motors

Operate on DC voltage and varying the voltage can easily control their speed.

BRUSHED

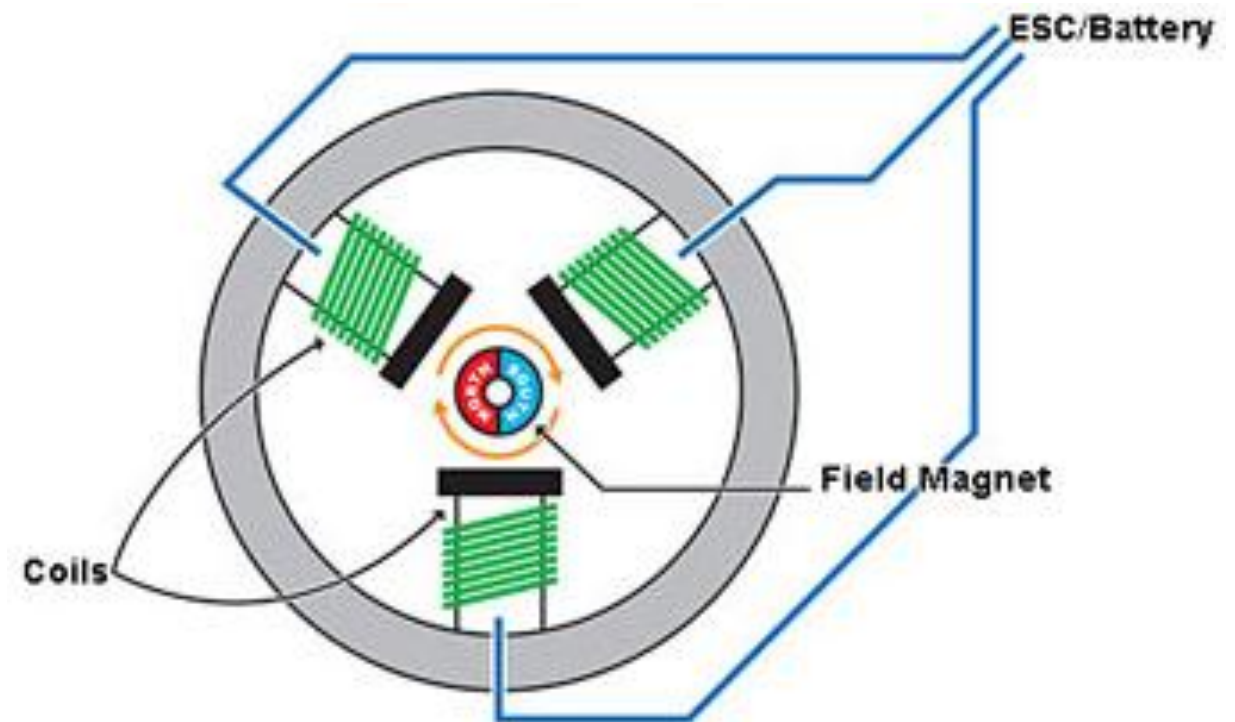
- In graphite/wear
- Limits on the max rotational velocity / contact pressures
- Glints during communications and disturbs
- Heavy rotor due to winding
- Hard to dissipate internal heat



- Applications: starter motors, fan motors, windshield wiper motors, etc.
- Disadvantages: More costlier, require more maintenance compared to AC motors.

BRUSHLESS

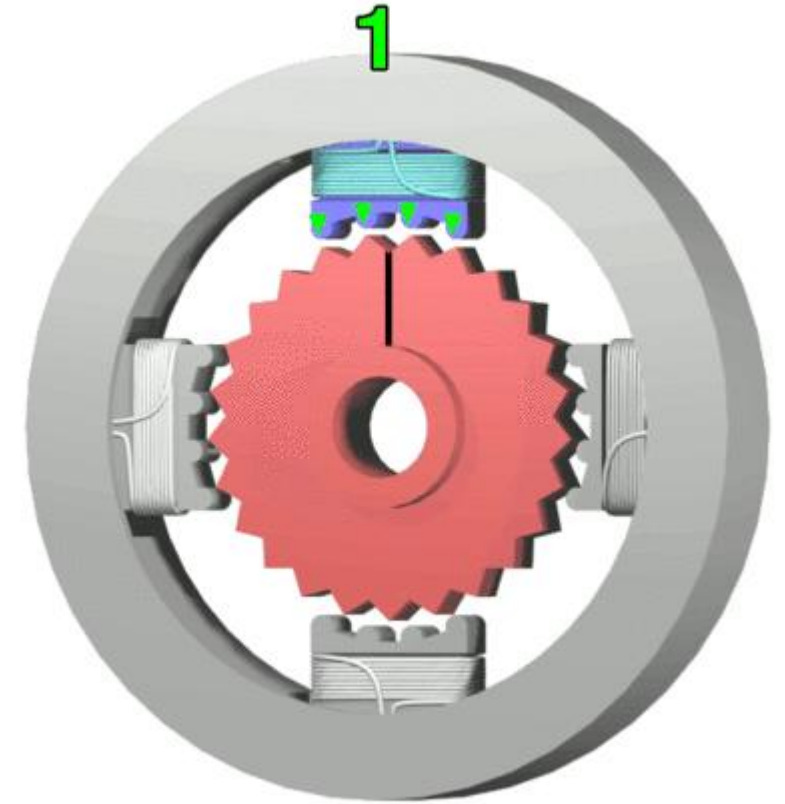
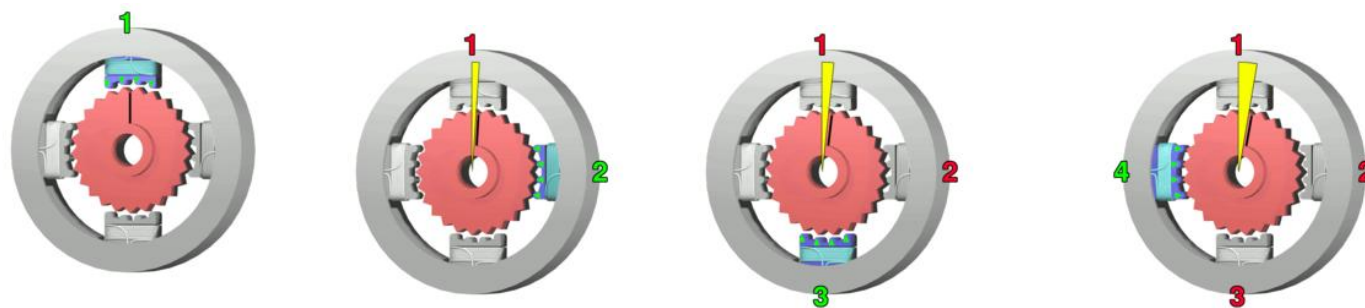
- Exchange of the rotor and stator (winding externally; rotational permanent magnet internally)
- Electronic commutator on the base of an encoder (servo motors)



Electromechanical actuators

Stepper motors

Brushless DC electric motor that divides a full rotation into a number of equal steps. The motor's position can then be commanded to move and hold at one of these steps without any feedback sensor (an open-loop controller), as long as the motor is carefully sized to the application.



Frame 1: The top electromagnet (1) is turned on, attracting the nearest teeth of the gear-shaped iron rotor. With the teeth aligned to electromagnet 1, they will be slightly offset from right electromagnet (2).

Frame 2: The top electromagnet (1) is turned off, and the right electromagnet (2) is energized, pulling the teeth into alignment with it. This results in a rotation of 3.6° in this example.

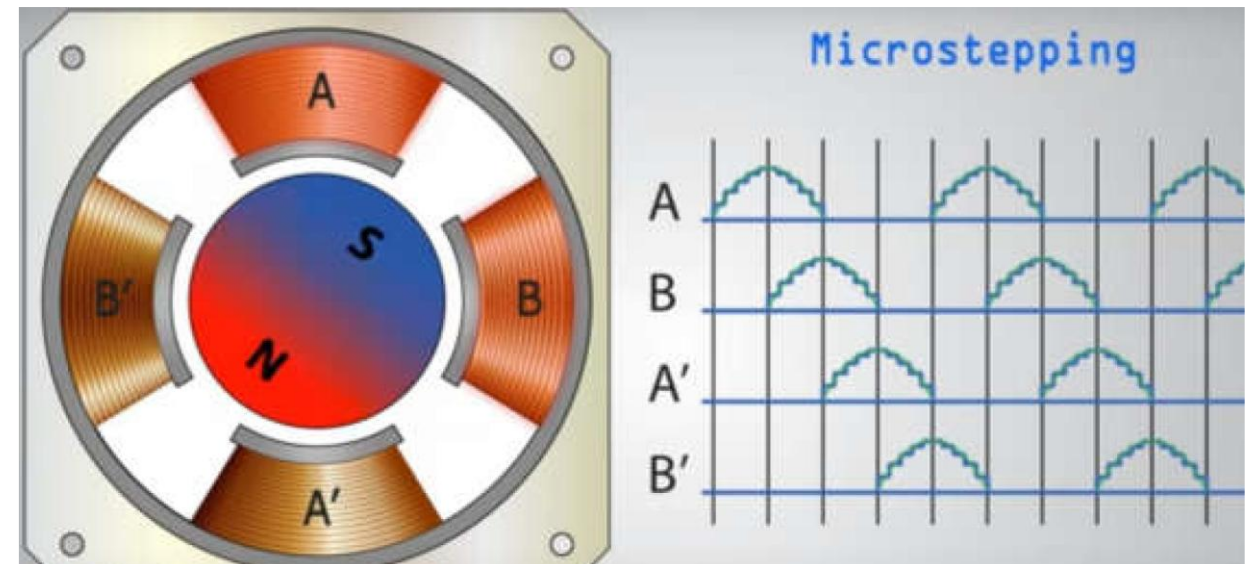
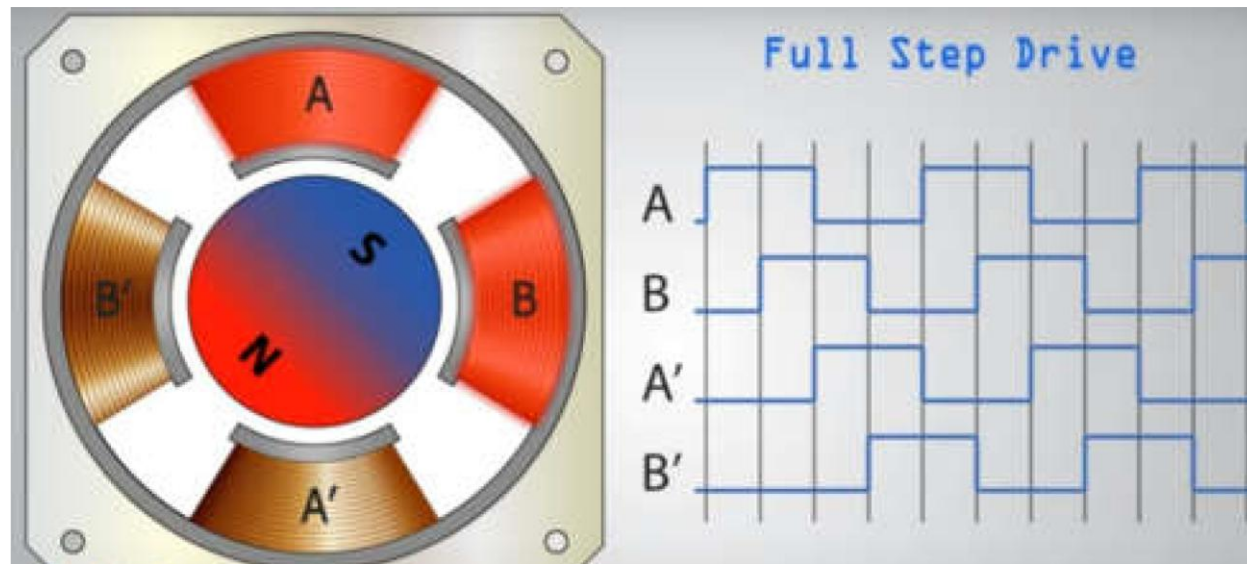
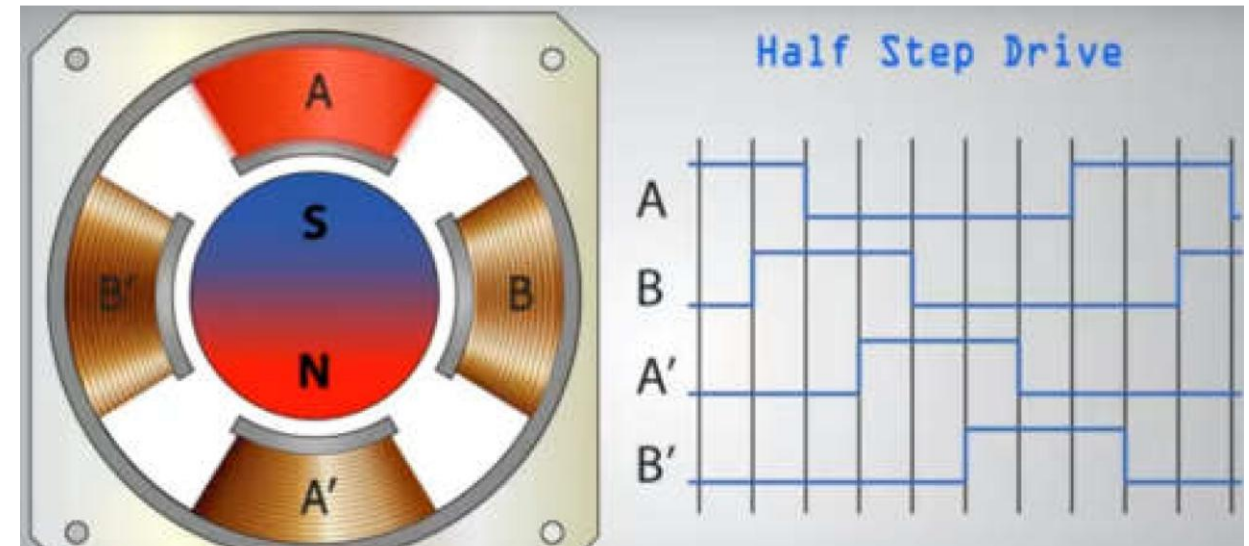
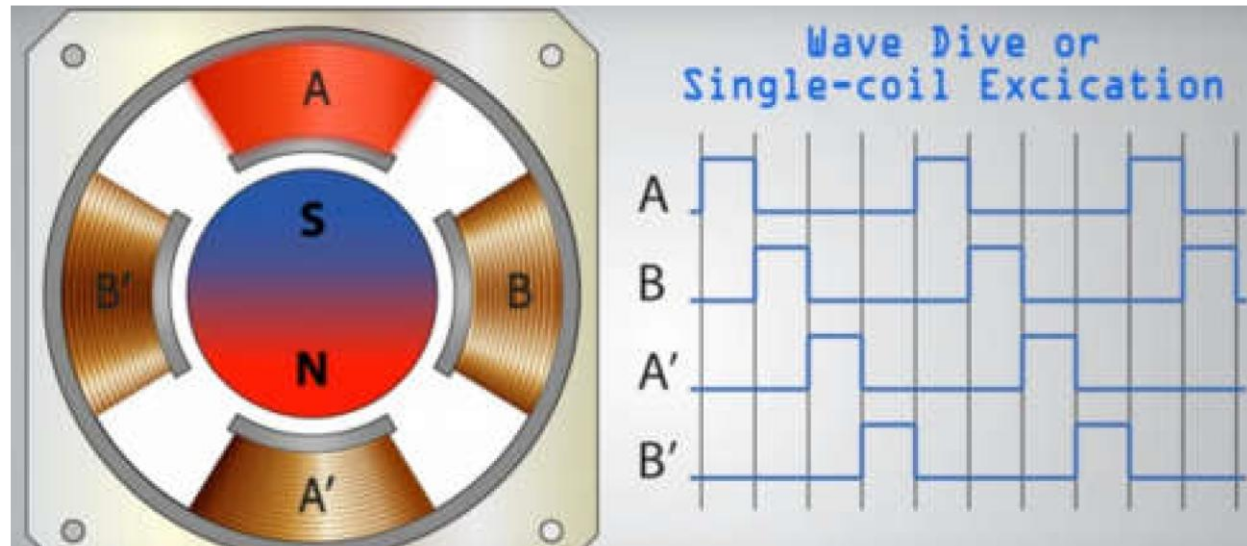
Frame 3: The bottom electromagnet (3) is energized; another 3.6° rotation occurs.

Frame 4: The left electromagnet (4) is energized, rotating again by 3.6° . When the top electromagnet (1) is again enabled, the rotor will have rotated by one tooth position; since there are 25 teeth, it will take 100 steps to make a full rotation in this example.



Electromechanical actuators

Stepper motors



Electromechanical actuators

AC motors

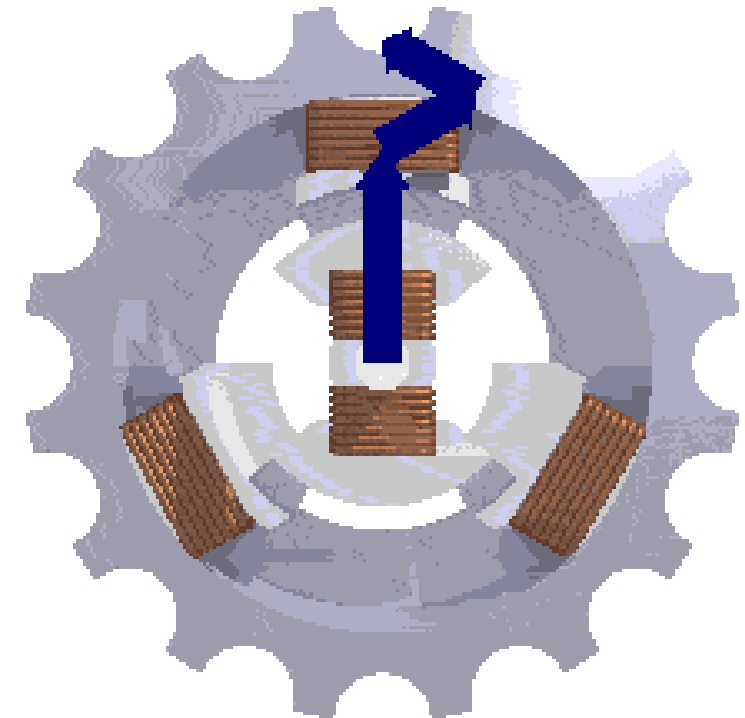
An AC motor is an electric motor driven by an alternating current (AC).

It commonly consists of two basic parts, an outside stationary stator having coils supplied with alternating current to produce a rotating magnetic field, and an inside rotor attached to the output shaft that is given a torque by the rotating field.

AC motors are the most popular since they use standard AC power, do not require brushes and commutator, and are therefore less expensive.

Synchronous motors

A synchronous electric motor is an AC motor in which, at steady state, the rotation of the shaft is synchronized with the frequency of the supply current; the rotation period is exactly equal to an integral number of AC cycles. Synchronous motors contain electromagnets on the stator of the motor that create a magnetic field which rotates in time with the oscillations of the line current. The rotor turns in step with this field, at the same rate



Electromechanical actuators

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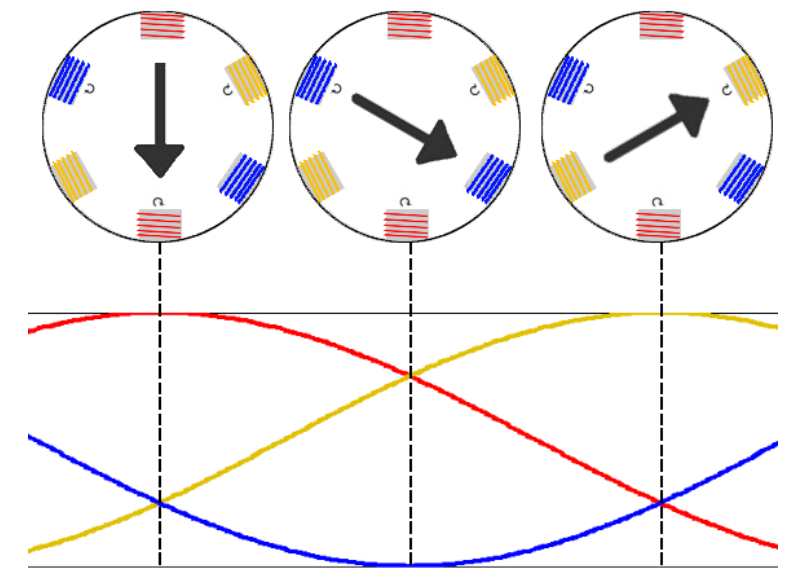
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Induction (asynchronous) motors

An induction motor's rotor rotates at a slower speed than the stator field. The induction motor stator's magnetic field is therefore changing or rotating relative to the rotor.

This induces an opposing current in the induction motor's rotor, in effect the motor's secondary winding, when the latter is short-circuited or closed through an external impedance. **The rotating magnetic flux induces currents in the windings of the rotor which in turn creates magnetic fields in the rotor that react against the stator field.**

The cause of induced current in the rotor windings is the rotating stator magnetic field, so to oppose the change in rotor-winding currents the rotor will start to rotate in the direction of the rotating stator magnetic field. The rotor accelerates until the magnitude of induced rotor current and torque balances the applied load.



Electromechanical actuators

AC motors

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Induction (asynchronous) motors



Special Thanks
Sajith K V

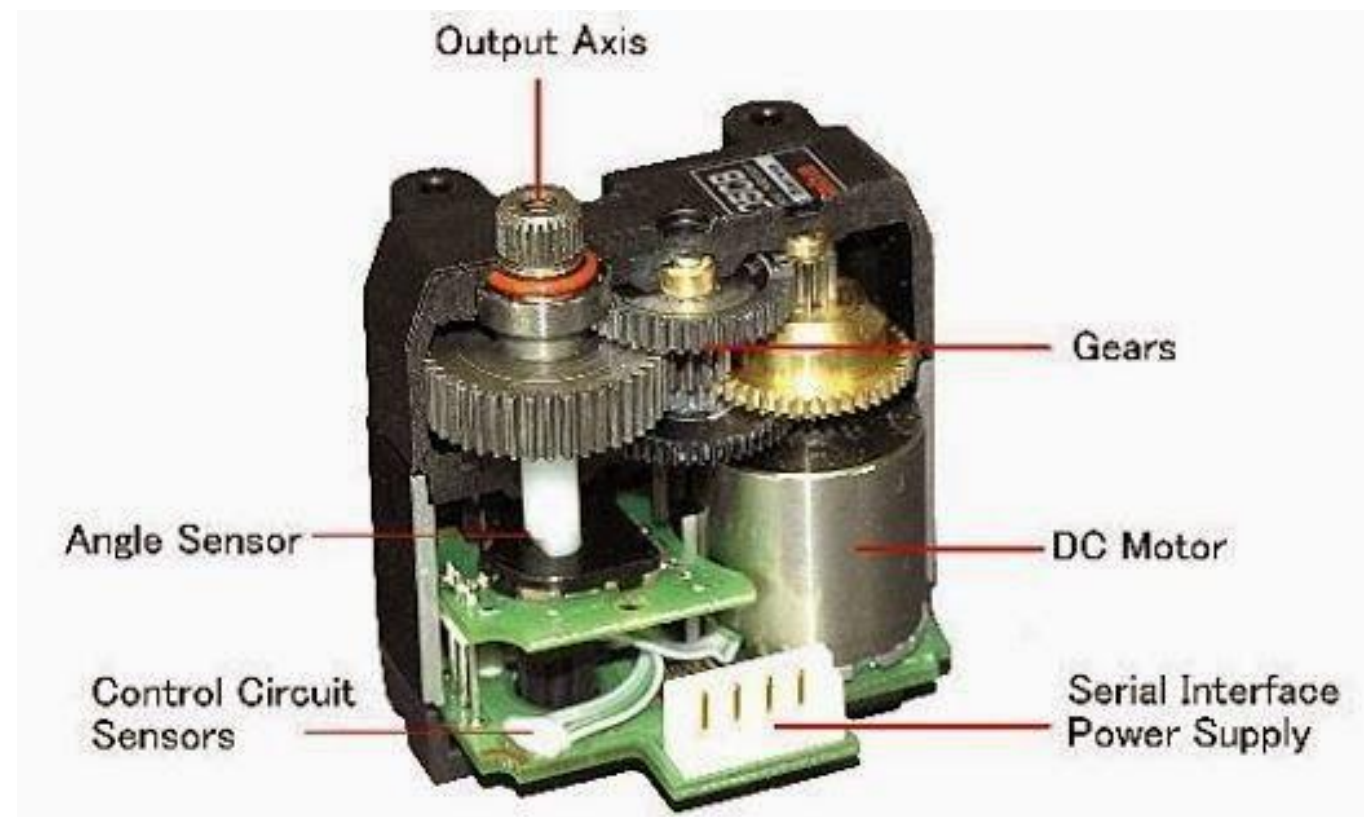
www.LearnEngineering.org



Servomotors

A servomotor is composed of a DC, AC or an AC induction motor and a feedback control

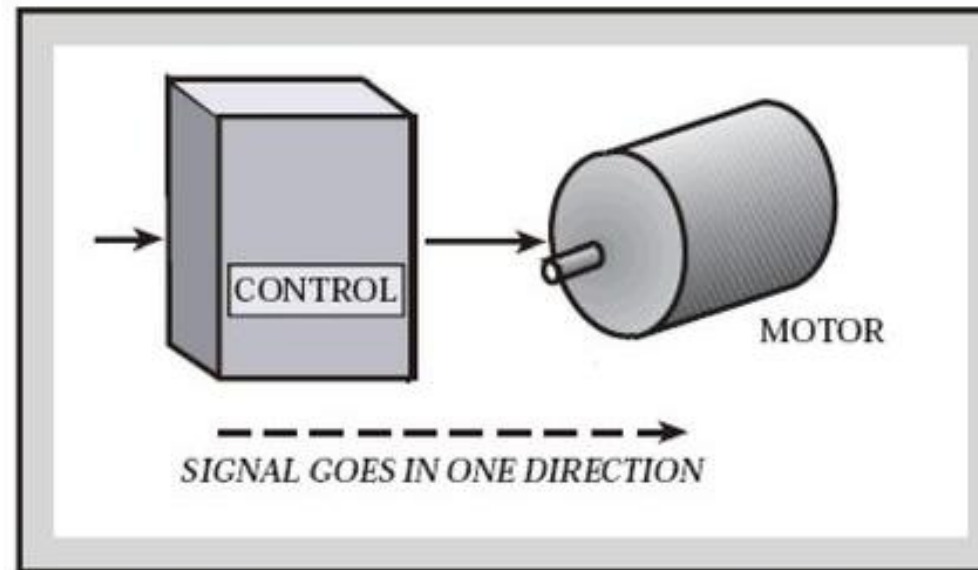
- Control angle NOT speed
- High torque
- Only 180 degree rotation



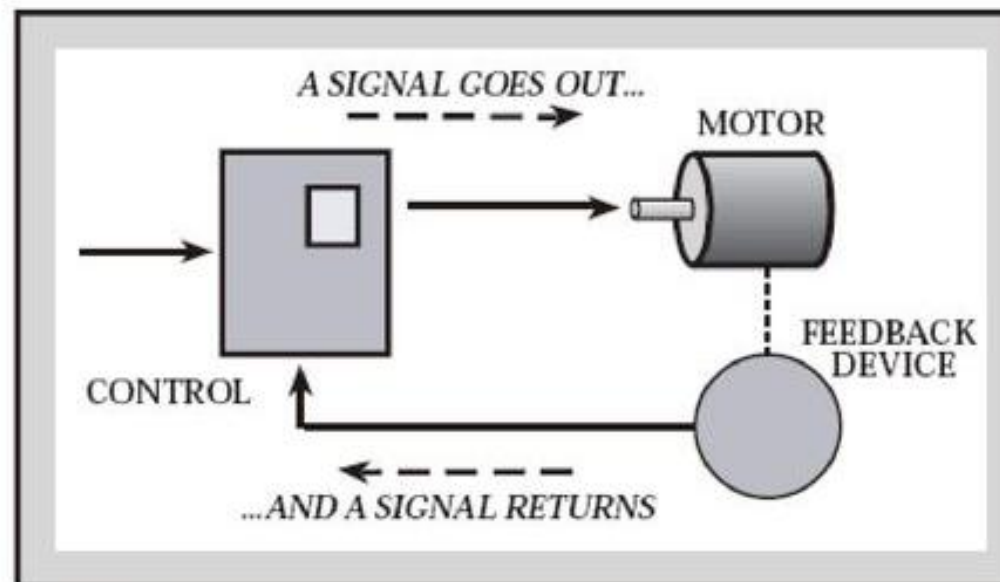
Servomotors

A servomotor is a closed-loop system: it is constantly taking note of its location and sending that information to the motor to determine if it should keep spinning

Open-loop

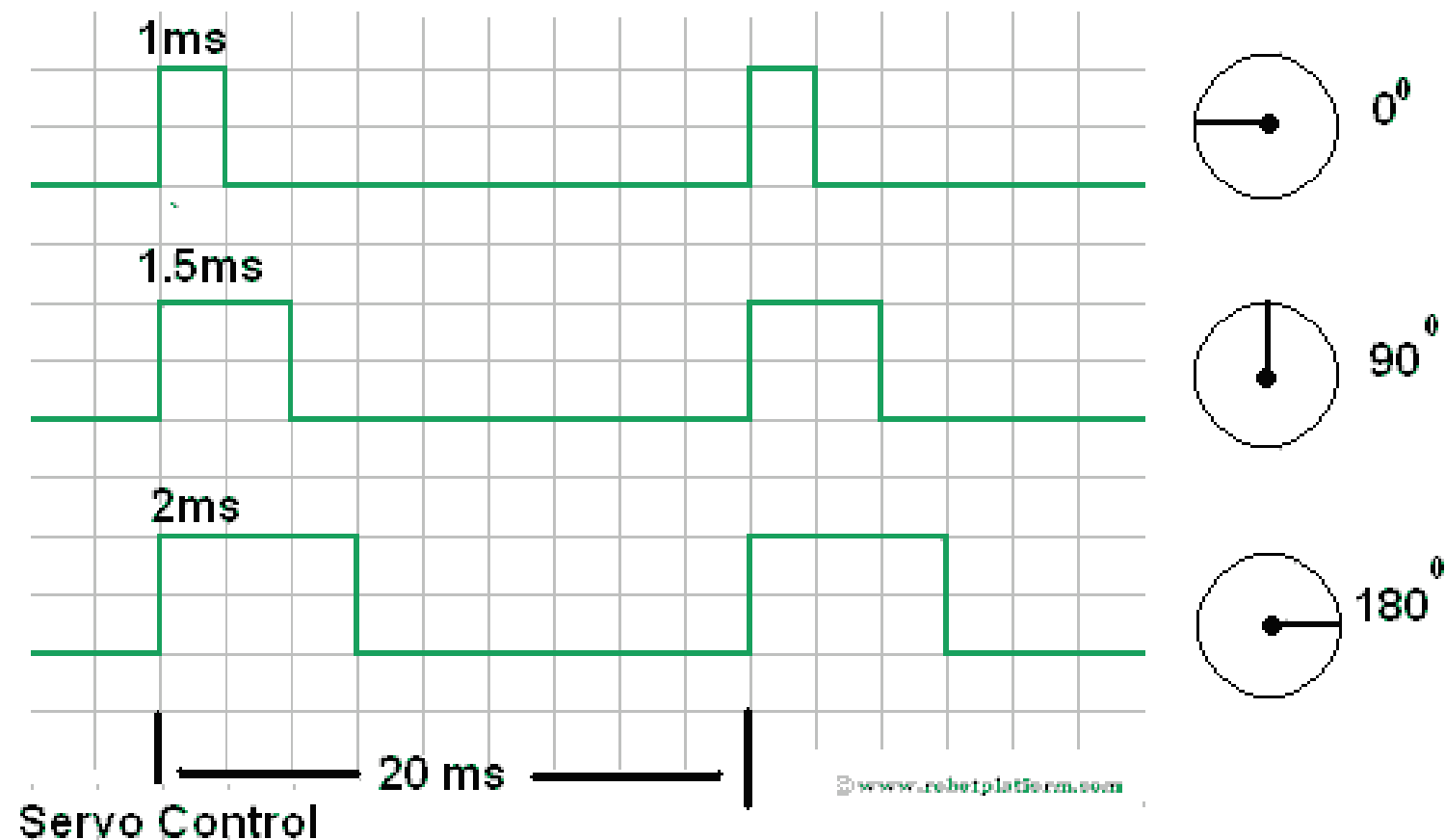


Closed-loop

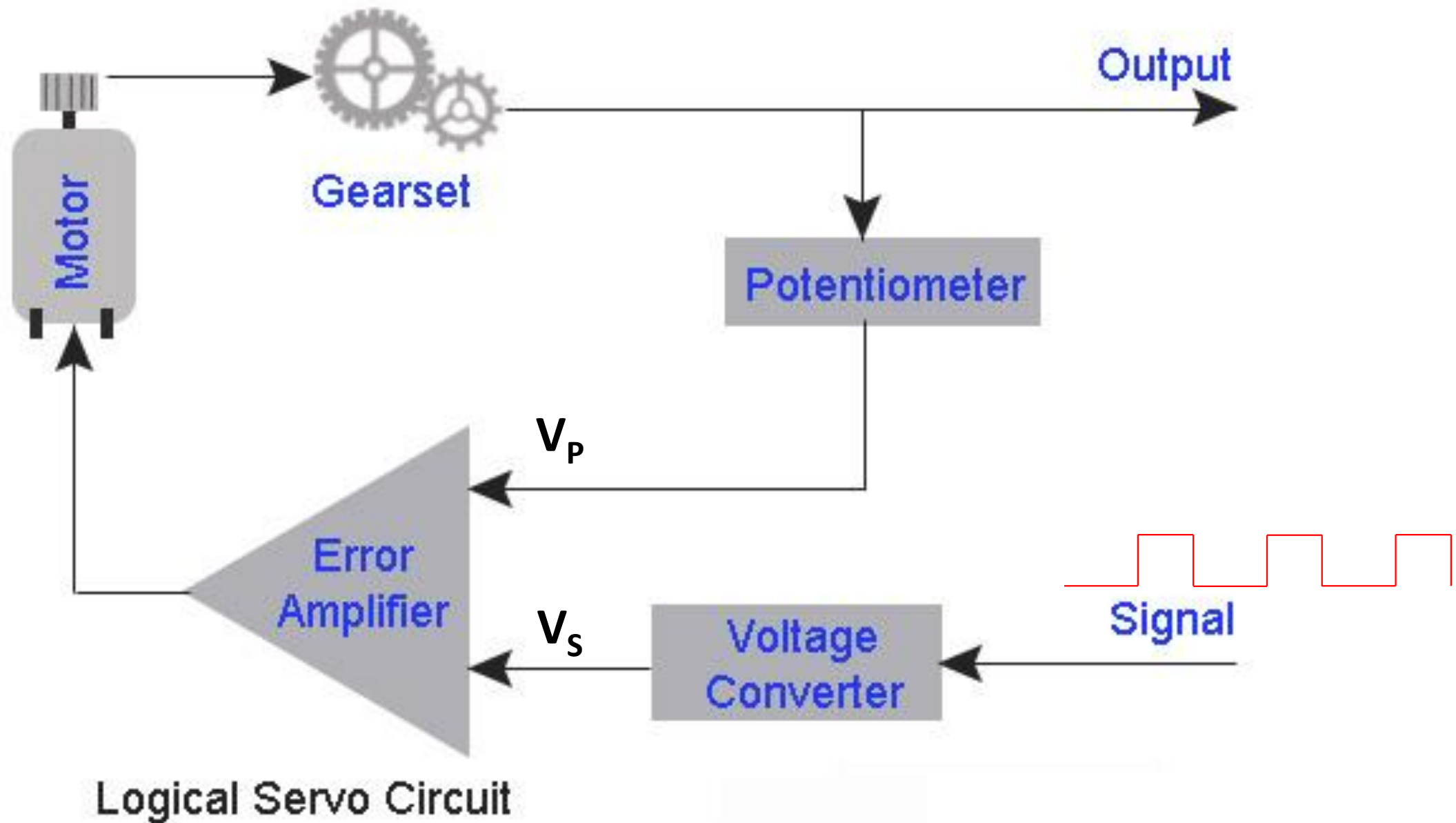


Servomotors

- DC servomotors are controlled with a PWM signal (square wave)
- In order to accurately hold a position servos need a pulse every 20ms
- The duration of the pulse decides how far the servo rotates



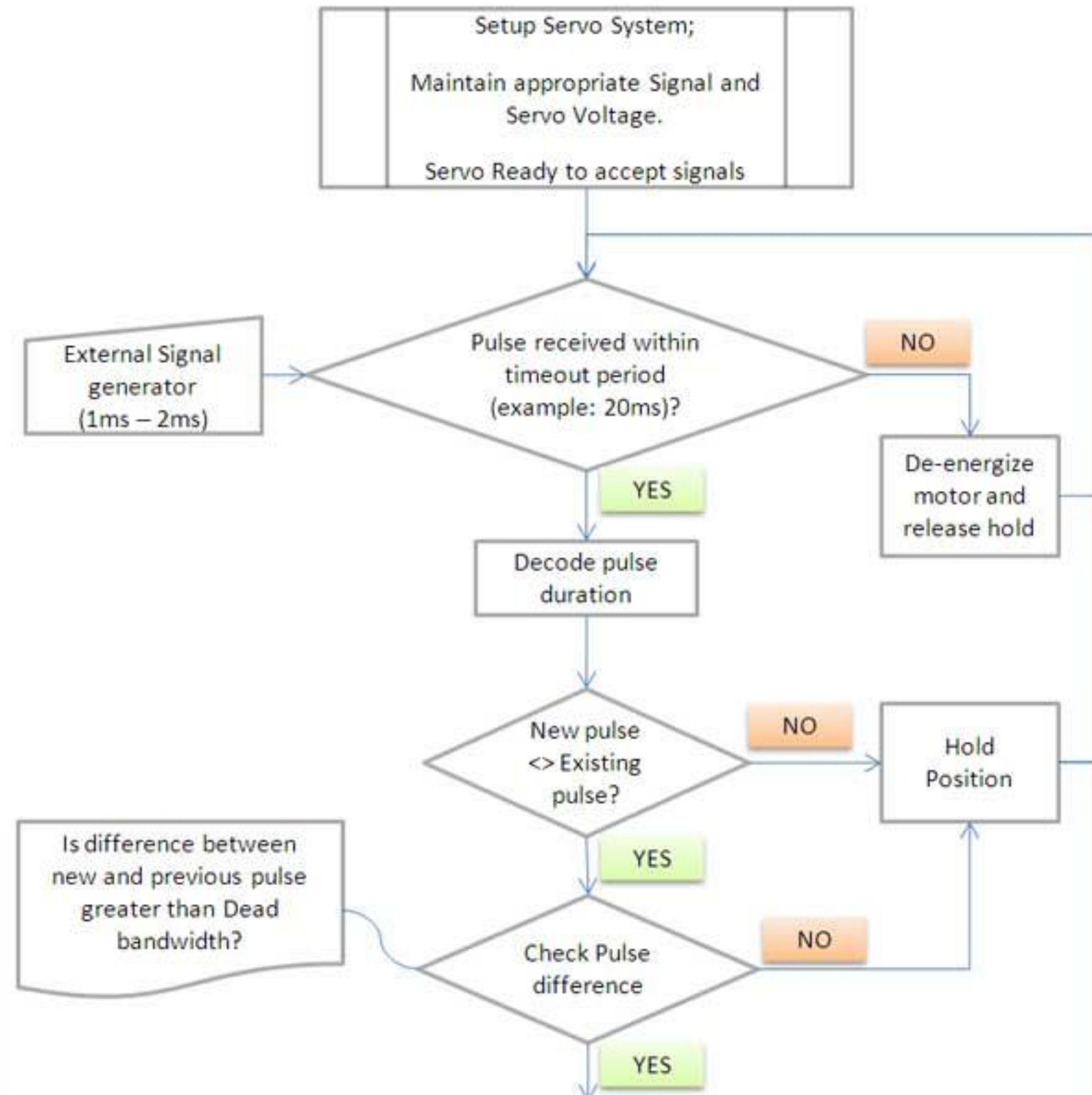
Servomotors



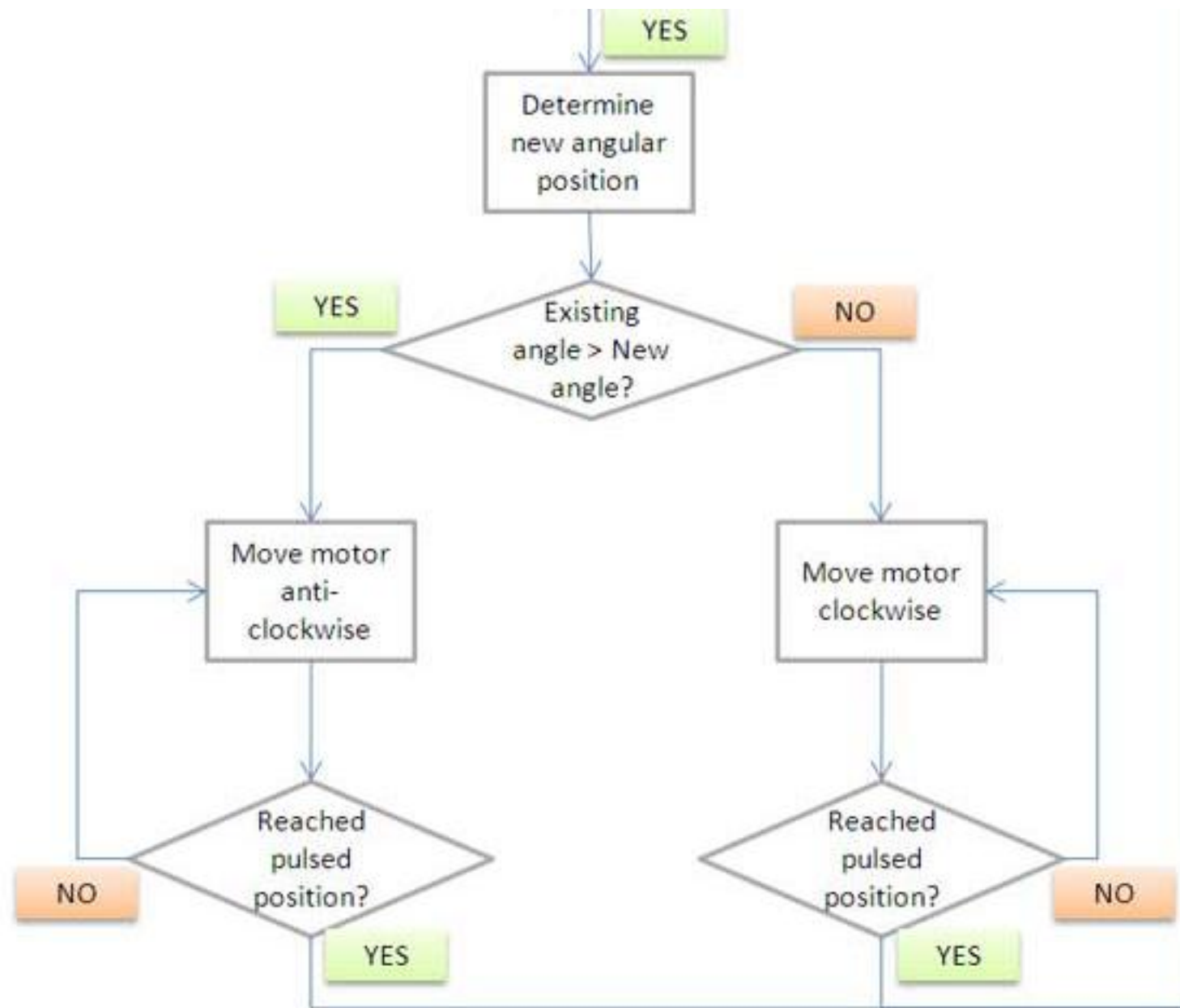
- $V_P < V_S$ rotation to decrease angle
- $V_P = V_S$ keep position
- $V_P > V_S$ rotation to increase angle



Servomotors

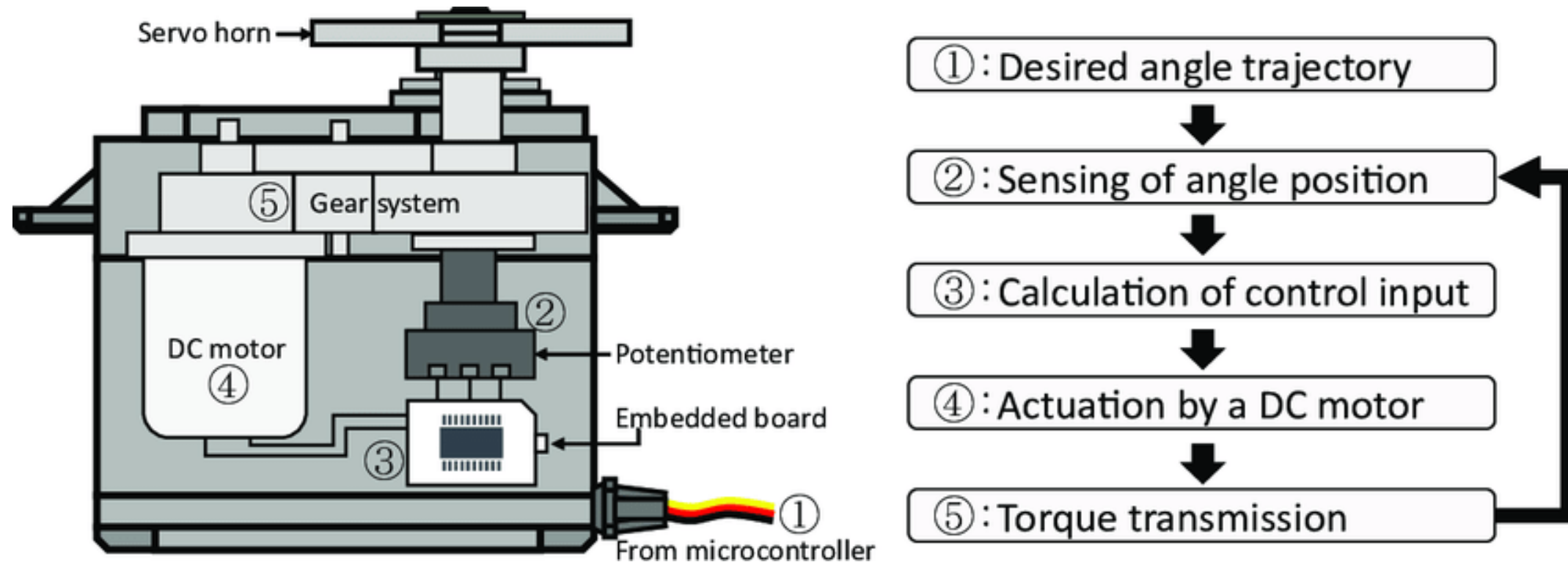


Servomotors



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Ground = brown or black

Power = red

Control = white, orange or yellow

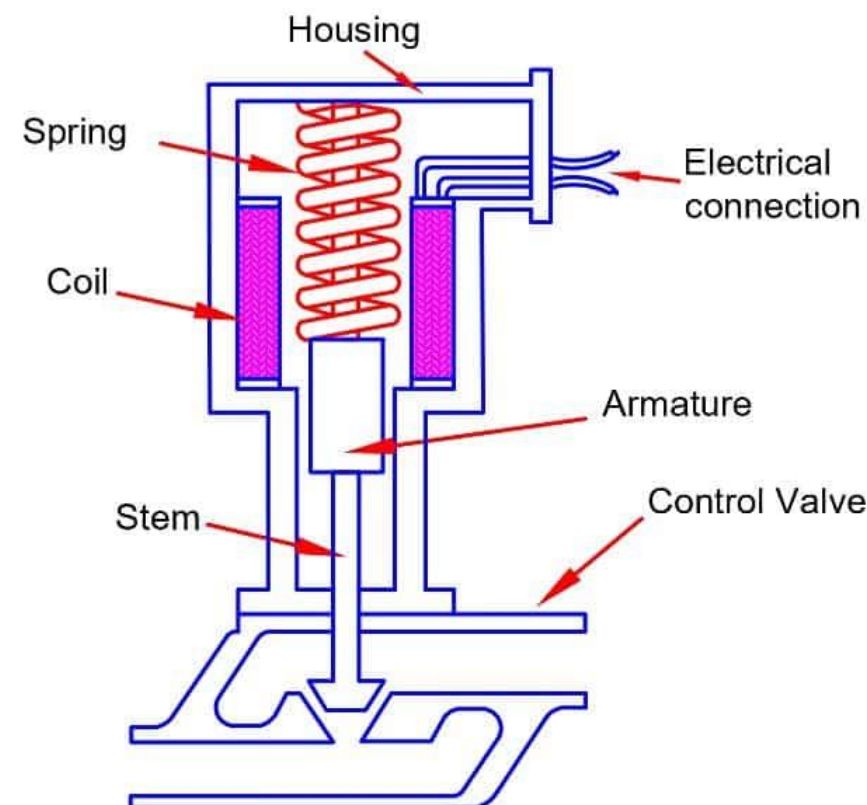


Electromagnetic actuators

Solenoids

Electromechanical solenoids consist of an **electromagnetically inductive coil**, wound around a movable steel or iron slug (termed the armature). The coil is shaped such that the armature can be moved in and out of the center, altering the coil's inductance and thereby **becoming an electromagnet**.

This actuator converts an electrical current into a proportional force.



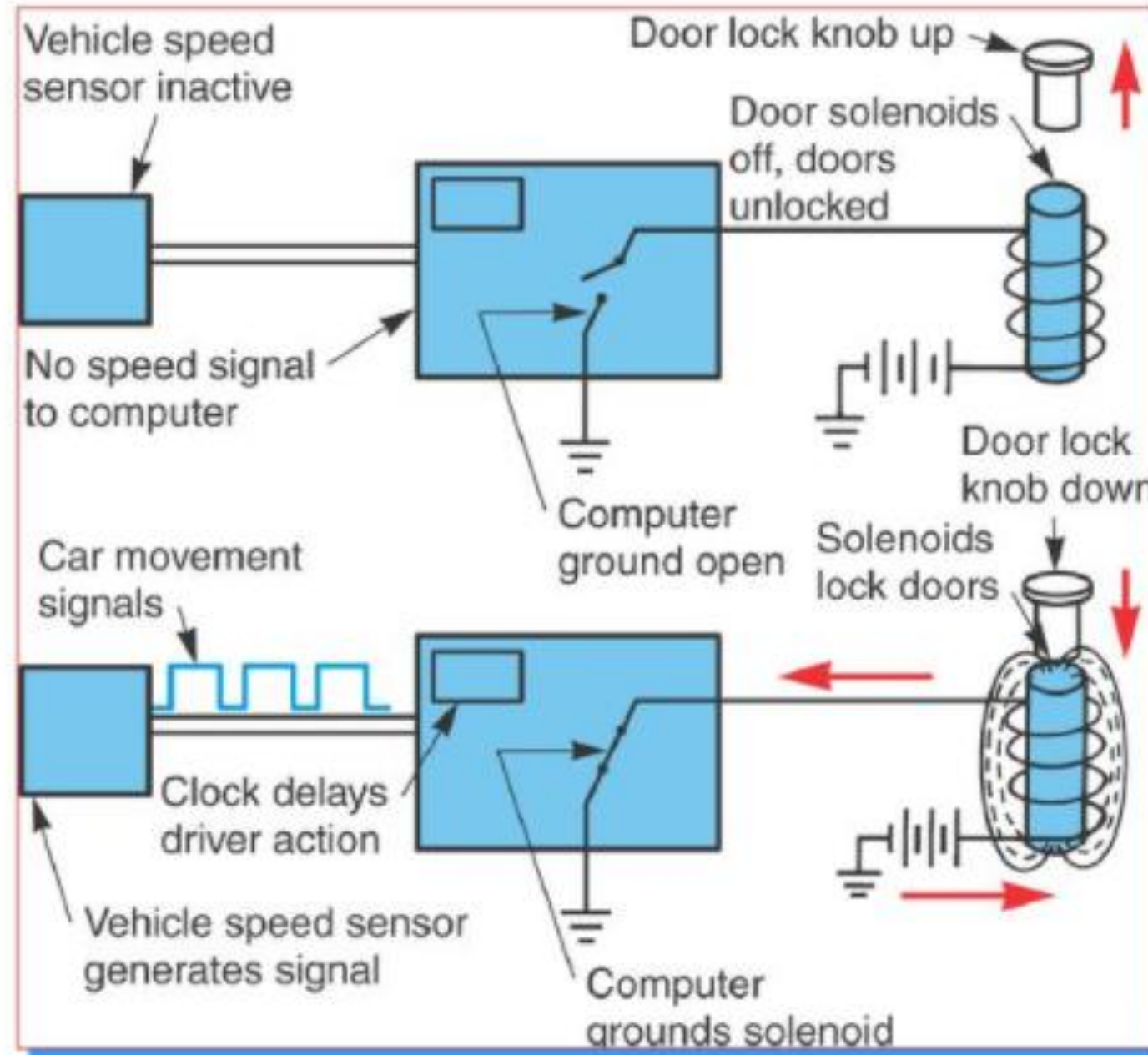
By applying a current I , the magnet is pushed or pulled with a force F . The direction of motion is always increasing the magnetic coupling and the inductance.

Plunger movement is used to operate a device (fuel injectors, vacuum valves, door locks...)



Electromagnetic actuators

Solenoids

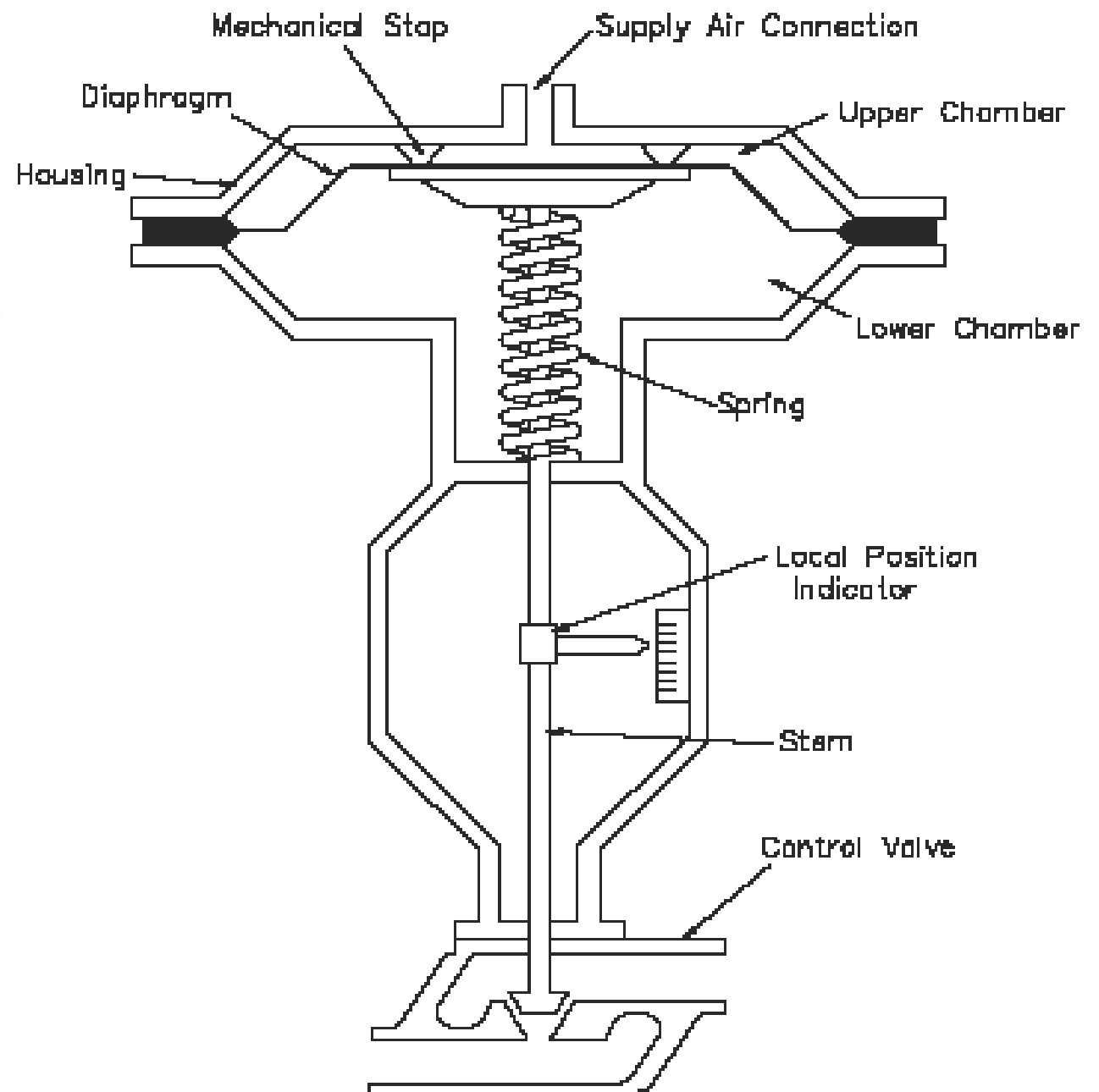
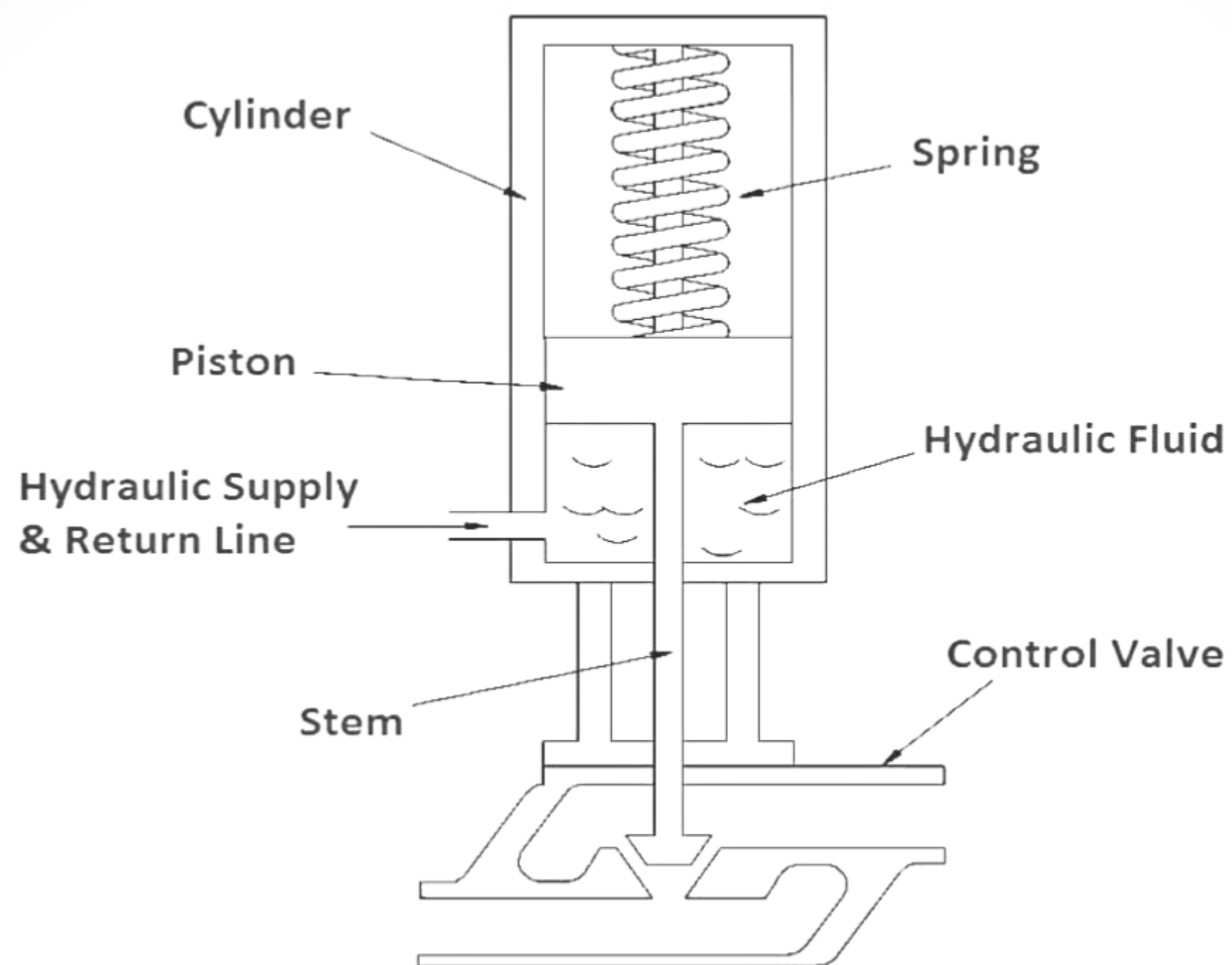


Automatic door locks



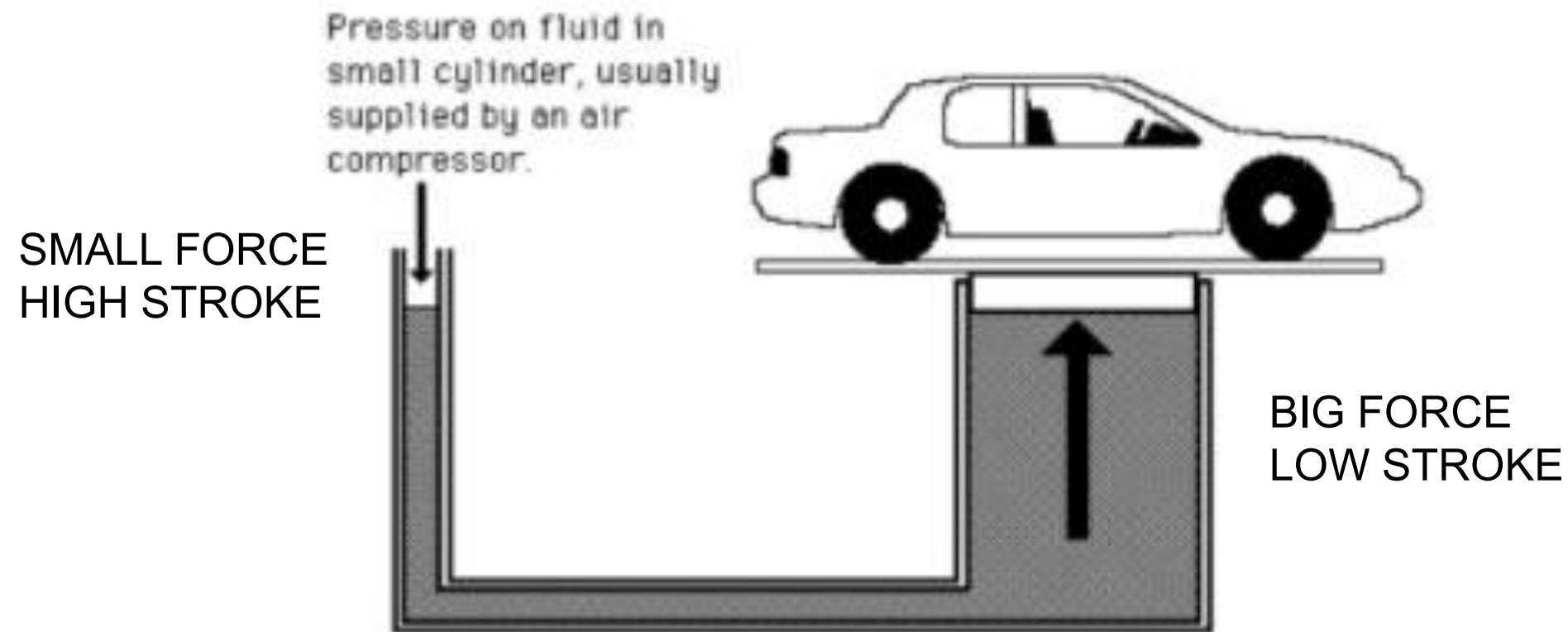
Hydraulic and Pneumatic actuators

- Hydraulic and pneumatic actuators are normally either rotary motors or linear piston/cylinder or control valves
- Ideally suited for generating very large forces coupled with large motion in a cost-effective manner



Hydraulic actuators

The basic rule of using hydraulic power is Pascal's principle: pressure exerted on a fluid is distributed equally throughout the fluid

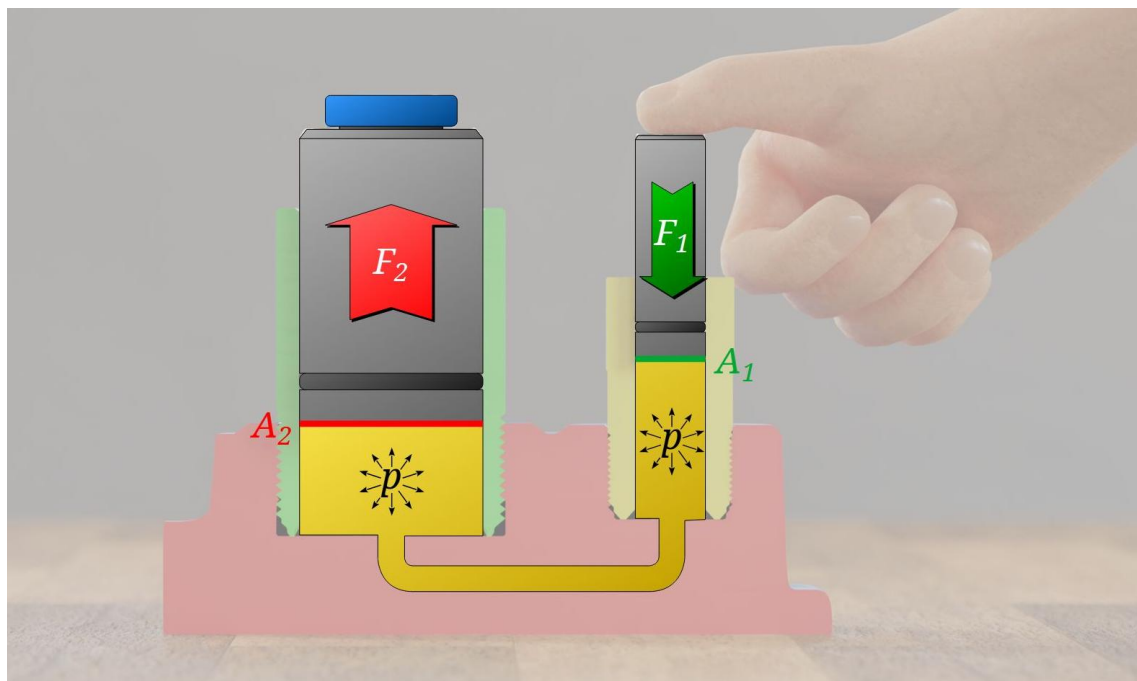


Hydraulics uses incompressible liquids so the applied pressure from one end (small arrow) is equal to the desired pressure on the other end (big arrow)

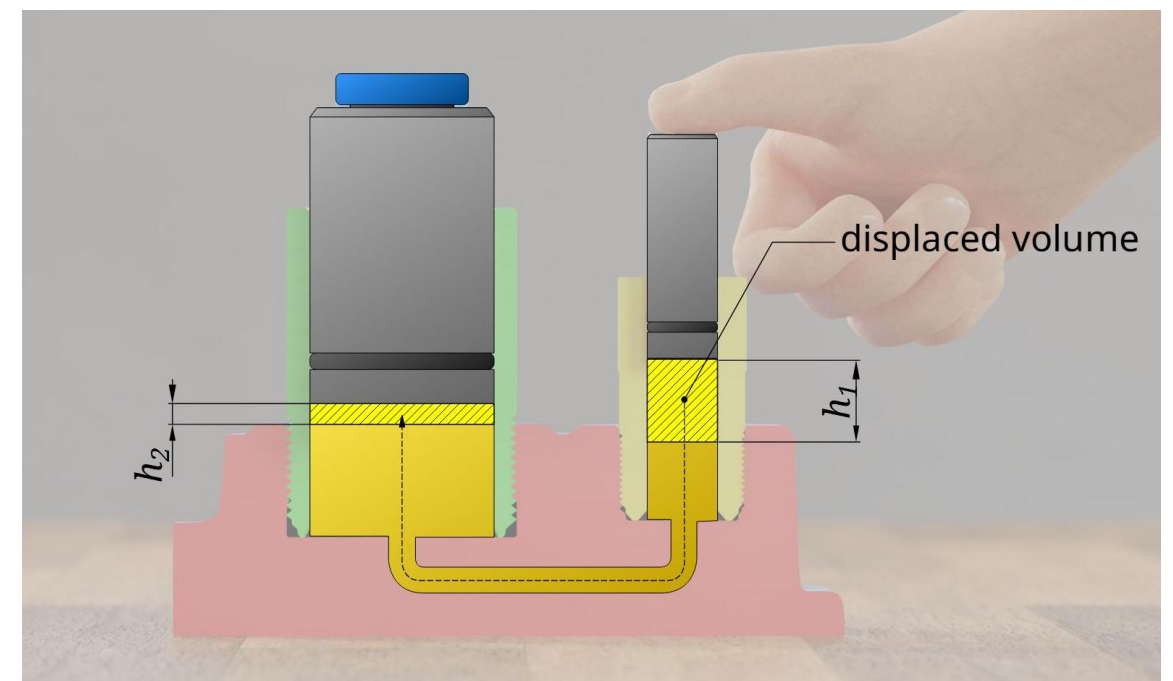


Hydraulic actuators

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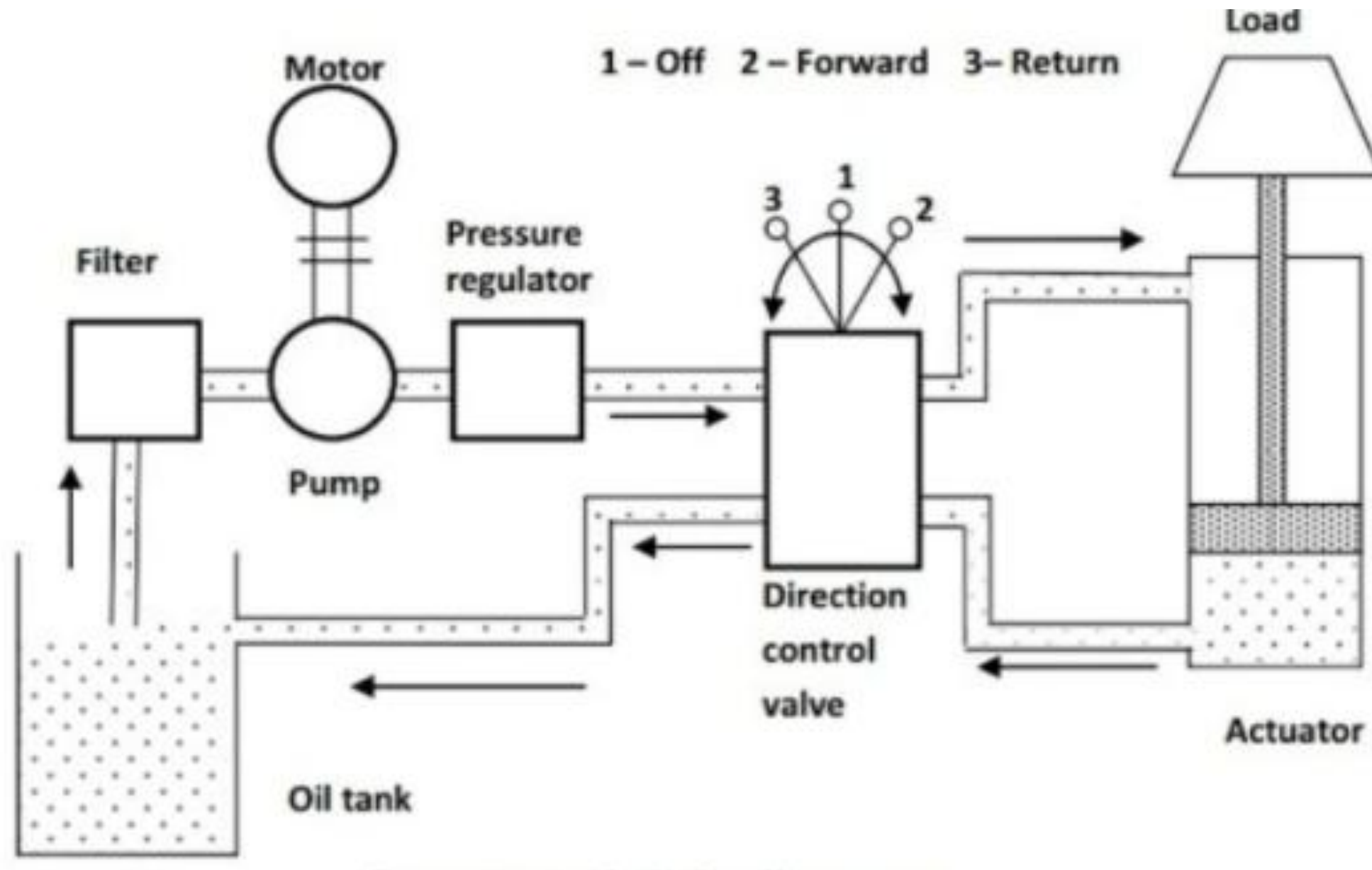
$$F_2 = pA_2 = F_1 \frac{A_2}{A_1}$$



$$h_2 = \frac{V}{A_2} = h_1 \frac{A_1}{A_2}$$



Hydraulic system



Hydraulic systems are used to control and transmit power. A pump driven by prime mover (electric motor) creates flow of fluid

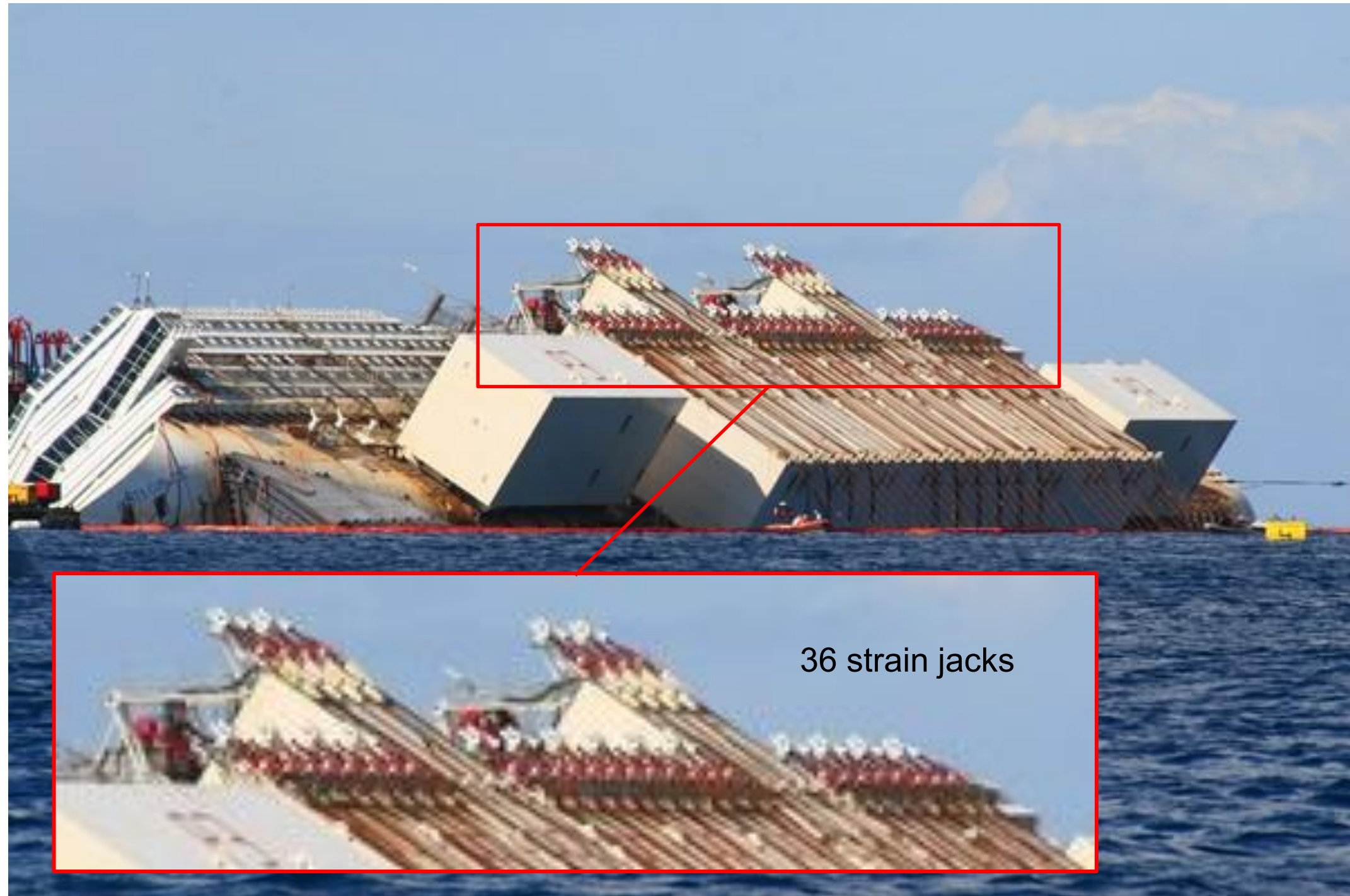


Hydraulic actuators

- Hydraulic jack
- Hydraulic brakes
- Hydraulic ram
- High precise positioning for heavy loads



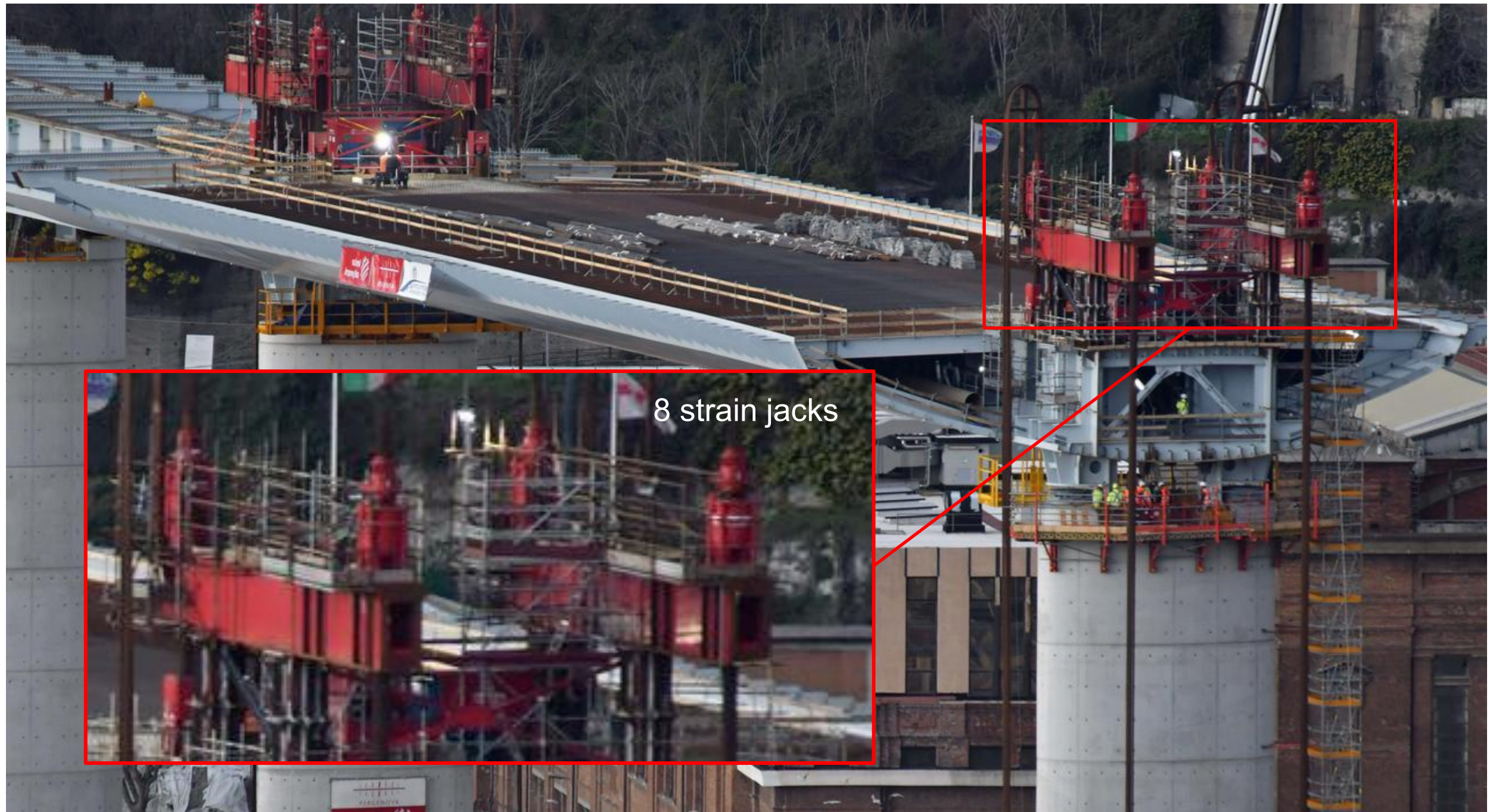
Hydraulic actuators



Costa Concordia shipwreck



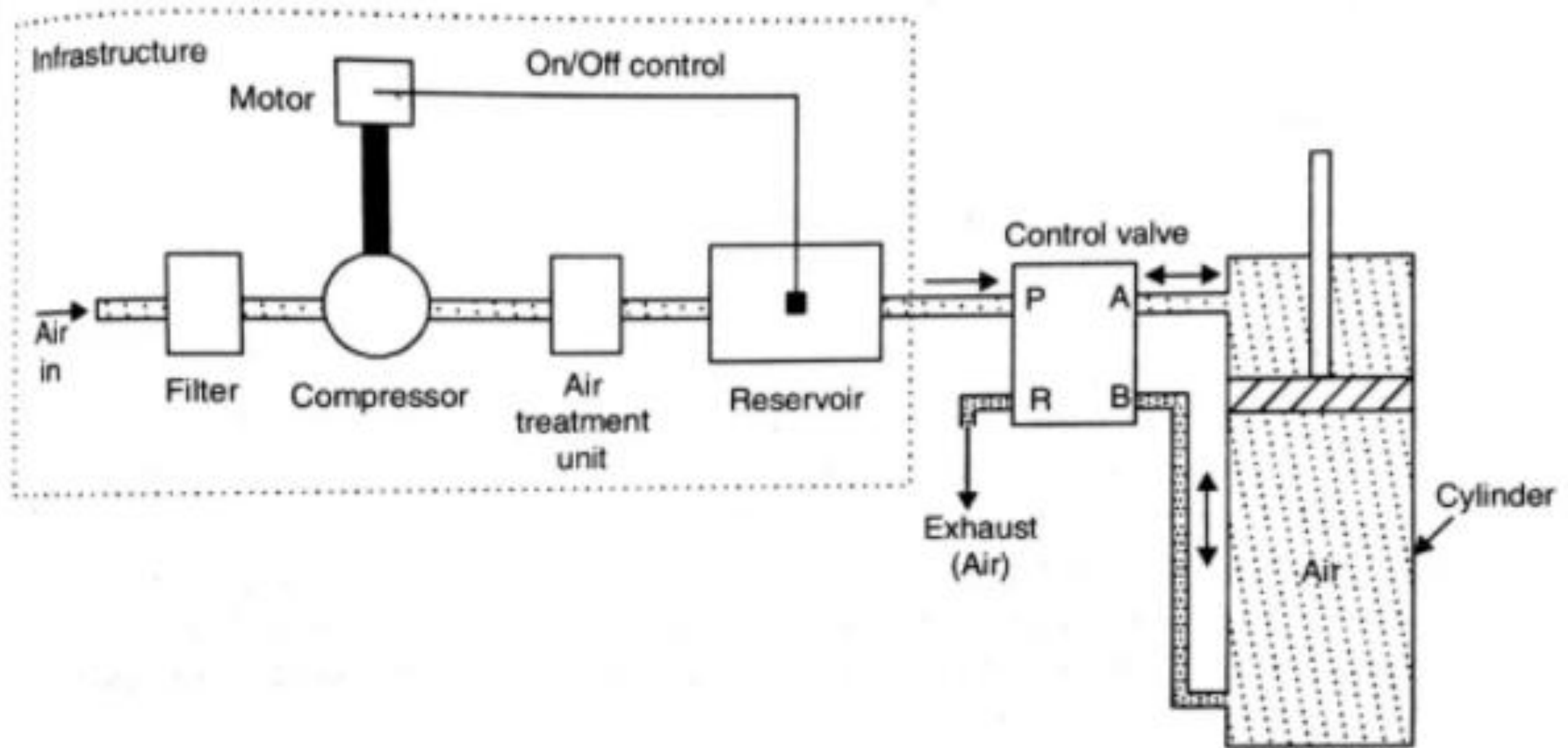
Hydraulic actuators



Morandi bridge –
Genova San Giorgio



Pneumatic system



Pneumatic systems use pressurized air to transmit and control power



Hydraulic vs Pneumatic actuators

	HYDRAULIC	PNEUMATIC
FLUID	Pressurized liquid	Compressed gas
PRESSURE RANGE	Up to 700 bar	Up to 10 bar
SYSTEM	Closed system	Open system
LEAKAGE	Slows down when occurs	Does not affect the system much
VALVES	Operations are difficult	Operations are easy
WEIGHT	Heavier	Lighter
FLUIDIC SOURCE	Pumps	Compressors
FIRE HAZARDS	Unsafe	Free
LUBRICATION	Automatic	Special arrangements needed
COST	Expensive	Cheap
SPEED	Low	High
NOISE	High	High
POSITIONING SPEED	High	Poor
EFFICIENCY	Higher	Lower



Take home lesson

Traditional electromagnetic motors are very commonly used for their:

- versatility**
- robustness**
- reliability...**

...but

New technological approaches in robotics and new application scenarios may require other characteristics

